

CHANGSHA HUANGHUA INTL, China

WMO: 592871

Lat: 28.189N

Lon: 113.220E

Elev: 217

StdP: 14.58

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.5	30.5	17.7	13.5	45.3	22.7	17.2	42.6	16.7	41.2	15.6	40.2	4.4	340	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	98.8	80.0	97.0	79.8	95.1	79.6	83.6	93.1	82.5	91.5	81.4	90.0	7.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	162.1	88.1	80.4	159.3	87.6	79.0	151.9	86.2	47.7	93.7	46.4	91.6	45.3	90.3	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	14.8	13.0	DB	24.8	102.4	2.5	1.9	23.0	103.8	21.6	104.9	20.2	106.0	18.4	107.4
			WB	24.1	84.3	2.5	1.9	22.3	85.7	20.8	86.9	19.4	88.0	17.6	89.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	41.9	46.3	55.4	65.9	73.9	80.1	86.1	84.8	77.8	68.1	57.0	46.3	(6)
(7)		DBStd	16.31	6.36	8.38	8.21	7.41	5.56	5.06	4.32	4.96	6.25	5.81	7.19	5.47	(7)
(8)		HDD50	618	262	159	37	2	0	0	0	0	0	20	138		(8)
(9)		HDD65	2506	716	526	314	76	4	0	0	0	2	35	253	580	(9)
(10)		CDD50	6243	11	56	205	481	742	903	1118	1080	834	560	230	23	(10)
(11)		CDD65	2656	0	3	17	104	281	453	653	615	387	130	13	0	(11)
(12)		CDH74	28023	2	29	155	803	2169	4454	8532	7423	3517	841	98	0	(12)
(13)		CDH80	13110	0	5	34	239	817	1906	4562	3766	1528	238	15	0	(13)
(14)	Wind	WSAvg	5.6	5.6	5.9	5.4	5.6	5.4	4.9	6.1	5.8	5.6	5.7	5.4	5.5	(14)
(15)	Precipitation	PrecAvg	59.4	3.0	3.5	6.0	7.3	8.4	9.1	5.9	4.8	3.2	2.5	3.4	2.2	(15)
(16)		PrecMax	71.4	7.1	7.6	14.5	10.8	17.8	19.4	15.4	13.3	9.6	6.1	8.1	5.5	(16)
(17)		PrecMin	40.0	0.6	0.5	3.6	3.2	2.2	4.4	0.4	1.1	0.1	0.1	0.5	0.0	(17)
(18)		PrecStd	8.9	1.6	1.6	2.4	2.3	3.4	3.6	3.9	2.6	2.2	1.4	2.4	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.1	78.6	84.4	89.8	93.6	97.0	100.8	101.8	98.4	91.3	82.0	69.5	(19)
(20)		MCWB	54.5	63.3	69.4	72.2	76.2	79.3	78.9	78.6	80.2	73.0	65.9	55.4		(20)
(21)		2%	DB	60.9	71.2	78.5	85.9	91.3	95.1	98.9	98.9	95.1	85.8	76.8	64.3	(21)
(22)		MCWB	48.9	59.0	64.8	70.5	74.3	79.2	79.7	80.4	78.5	69.6	63.3	52.0		(22)
(23)		5%	DB	55.7	64.5	73.3	82.5	88.0	93.0	97.1	96.9	91.7	82.3	71.9	59.2	(23)
(24)		MCWB	47.4	55.3	61.7	68.3	73.7	78.9	79.9	79.9	77.1	67.4	60.6	49.4		(24)
(25)		10%	DB	51.8	58.7	68.1	78.8	84.5	89.7	95.3	94.8	88.2	78.8	68.2	55.4	(25)
(26)		MCWB	45.7	51.6	59.4	67.3	72.7	77.7	79.9	79.8	75.1	66.0	59.6	47.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.7	65.7	73.4	76.8	80.6	83.7	85.3	85.0	83.6	76.7	70.3	58.7	(27)
(28)		MCDWB	66.6	76.1	80.5	87.1	89.3	92.6	94.6	94.7	92.6	82.8	77.3	66.2		(28)
(29)		2%	WB	51.9	60.6	67.2	73.4	78.2	82.0	83.6	83.6	81.3	73.4	66.5	54.3	(29)
(30)		MCDWB	57.6	69.3	75.9	81.4	86.5	90.0	93.1	93.3	89.7	80.2	71.3	61.4		(30)
(31)		5%	WB	48.8	56.5	63.9	71.0	76.3	80.4	82.4	82.4	79.5	70.9	63.8	51.4	(31)
(32)		MCDWB	54.0	61.8	70.0	78.4	83.9	87.9	91.2	91.8	87.4	77.2	68.5	56.1		(32)
(33)		10%	WB	45.9	53.0	60.8	69.1	74.6	79.1	81.3	81.3	78.1	68.9	61.5	49.5	(33)
(34)		MCDWB	49.7	58.0	66.6	75.8	81.5	86.6	90.0	89.9	85.2	75.3	66.6	53.2		(34)
(35)	Mean Daily Temperature Range	MDBR	11.3	12.3	14.1	15.7	14.9	13.4	15.2	15.5	15.3	15.6	15.1	14.2		(35)
(36)		5% DB	MCDBR	23.6	24.7	24.9	23.2	21.9	18.2	18.4	19.2	20.3	22.7	23.8	24.9	(36)
(37)		MCWBR	13.0	13.2	11.7	10.2	8.9	6.2	5.3	5.5	6.9	8.4	11.1	13.3	(37)	
(38)		5% WB	MCDBR	18.9	20.4	19.9	18.9	17.5	15.9	16.4	17.5	17.7	16.8	16.2	19.9	(38)
(39)		MCWBR	11.4	11.9	11.0	10.0	8.7	6.9	6.2	6.7	7.4	8.1	9.0	11.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.695	0.736	0.787	0.768	0.696	0.678	0.607	0.673	0.699	0.735	0.682	0.664		(40)
(41)		taud	1.495	1.459	1.408	1.481	1.646	1.698	1.874	1.698	1.622	1.500	1.566	1.563		(41)
(42)		Ebn at Noon	171	180	183	194	210	213	228	212	199	177	173	171		(42)
(43)		Edh at Noon	81	90	100	95	81	77	64	76	80	85	75	73		(43)
(44)	All-Sky Solar Radiation	RadAvg	542	646	856	1067	1225	1305	1677	1450	1217	961	756	628		(44)
(45)		RadStd	119	183	97	136	179	154	166	123	131	150	132	141		(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	+0.52	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page